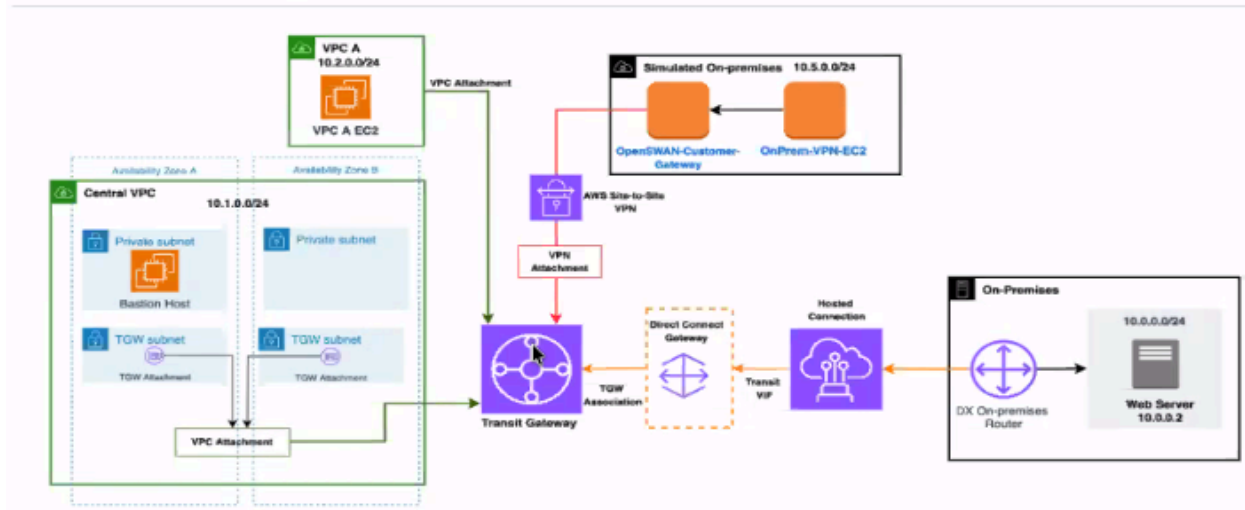


AWS Direct Connect - Transit Gateway Lab

In this lab I will be setting up a Direct connect Hosted Connection. I will create private virtual interfaces, (VIFs) to access resources in the VPCs, public virtual interfaces to reach public AWS services, and transit virtual interfaces to connect with Transit Gateways.

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Architecture Overview



1. Open the **AWS VPC Console**. Click **Transit Gateway Attachments** from the left side menu. Click **Create Transit Gateway attachment**. You will already see one TGW attachment for Central VPC. The new attachment we are creating is for connecting VPC A to Transit Gateway.

Transit gateways
Transit gateways
Transit gateway attachments
Transit gateway policy tables
Transit gateway route tables
Transit gateway multicast
Metering policies

Transit gateway attachments (2) [Info](#)

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resour...
☐	Central-VPC-TGW-At...	tgw-attach-049bdcf53cc26a62c	tgw-0daa30cea3f5a6163	Available	VPC
☐	transit-gateway-atta...	tgw-attach-05c36784d3f89891f	tgw-0daa30cea3f5a6163	Available	VPC

Select a transit gateway attachment

2. Enter a **Name tag**, select the **Transit Gateway ID** name (**Workshop-TGW**) and select the **VPC ID** as **VPC-A**. Select all the Availability Zones and subnets click **Create transit gateway attachment**.
3. VPC attachment should create successfully and will be in a **pending** state initially. It will take couple of minutes for attachment to become **Available**

Assign router for transit gateway for VPC A and Central VPC

1. On the left in the VPC Dashboard click on Route Tables
2. Select the check box for **VPC A Route Table**, scroll down select the **Routes** tab and click on **Edit routes**.
3. Add a route to Central VPC in the VPC A Private Route Table using a Destination of **10.1.0.0/24** with a Target as the transit gateway (TGW VPC A attachment).
4. Click on **Save routes**.

Edit routes

Destination	Target	Status	Propagated	Route Origin	
2600:1f18:785f:3300::/56	local	Active	No	CreateRouteTable	
	Q local				
10.2.0.0/24	local	Active	No	CreateRouteTable	
	Q local				
Q 10.1.0.0/24	Transit Gateway	Active	No	CreateRoute	Remove
	Q tgw-0daa30cea3f5a6163				

Add route

Cancel Preview Save changes

5. Similarly, add route for **10.2.0.0/24** with Target as Transit Gateway (TGW Central VPC Attachment) in Central VPC route table **Central-Private-RT**

Edit routes

Destination	Target	Status	Propagated	Route Origin	
2600:1f18:4e78:b900::/56	local	Active	No	CreateRouteTable	
	Q local				
10.1.0.0/24	local	Active	No	CreateRouteTable	
	Q local				
Q 10.2.0.0/24	Transit Gateway	Active	No	CreateRoute	Remove
	Q tgw-0daa30cea3f5a6163				

Add route

Cancel Preview Save changes

TGW Test Connectivity

1. Open the **EC2 Console**, and choose **Instances** from the side menu.
2. Click on the **Bastion-Host** instance and click the **Connect** button.

EC2 > Instances

Instances (1/5) Info Connect Instance state Actions

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check
☐	OnPrem-VPN-...	i-0f9ee0fd3095f3156	Running	t2.micro	2/2 checks passed
☐	OpenSWAN-C...	i-09162eb40a5dbfc4e	Running	t3.micro	3/3 checks passed
☐	VPC-A-Instance	i-0f7aba63c301140a3	Running	t2.micro	2/2 checks passed
☒	Bastion-Host	i-0e8d5a9d721587b5c	Running	t3.micro	3/3 checks passed
☐	OnPrem-DX-R...	i-01e5a3b2f23dc62b8	Running	m5.large	3/3 checks passed

3. In the Connect to instance page, select the **Session Manager** tab and click on **Connect** button.

EC2 Instance Connect **SSM Session Manager** SSH client EC2 serial console

SSM agent status

Ping status Online	Last ping time Thu Mar 12 2026 05:51:06 GMT-0500 (Central Daylight Time)	Session Manager connection status Connected	Agent version 3.3.3598.0
------------------------------	---	---	------------------------------------

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

Cancel Connect

4. Once connected, you can ping <VPC A EC2 Private IP>

```
sh-4.2$ ping 10.1.0.93
PING 10.1.0.93 (10.1.0.93) 56(84) bytes of data.
64 bytes from 10.1.0.93: icmp_seq=1 ttl=255 time=0.015 ms
64 bytes from 10.1.0.93: icmp_seq=2 ttl=255 time=0.025 ms
64 bytes from 10.1.0.93: icmp_seq=3 ttl=255 time=0.037 ms
64 bytes from 10.1.0.93: icmp_seq=4 ttl=255 time=0.023 ms
64 bytes from 10.1.0.93: icmp_seq=5 ttl=255 time=0.022 ms
64 bytes from 10.1.0.93: icmp_seq=6 ttl=255 time=0.024 ms
64 bytes from 10.1.0.93: icmp_seq=7 ttl=255 time=0.026 ms
64 bytes from 10.1.0.93: icmp_seq=8 ttl=255 time=0.024 ms
^C
--- 10.1.0.93 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7156ms
rtt min/avg/max/mdev = 0.015/0.024/0.037/0.007 ms
sh-4.2$
```

Creating a Direct Connect Gateway

First, create a Direct Connect gateway (DXGW) and associate it with the pre-existing Transit Gateway. This will allow the VPCs attached to the Transit Gateway to communicate through the Direct Connect gateway to the on-premises environment.

1. Open the **AWS Direct Connect** Console. Click **Direct Connect gateways** from the left side menu. Click **Create Direct Connect gateway**.
2. Enter a **Name**, and **Amazon side ASN** for Direct Connect gateway. You can use AS # **64512** as the BGP ASN for AWS side.

ASN (Autonomous System Number) is a unique identifier for BGP routing. We assign ASN 64512 to the Direct Connect Gateway and a separate ASN to Transit Gateway so they can exchange routes via BGP and establish connectivity between AWS and on-premises networks.

3. Finally, click **Create Direct Connect gateway** to create the gateway.

Direct Connect gateway settings

Name

A name to help you identify the new Direct Connect gateway.

Name must contain no more than 100 characters. Valid characters are a-z, 0-9, and hyphens (-).

Amazon-side ASN

The Autonomous System Number (ASN) for the new Direct Connect gateway.

Valid ranges are 64512 - 65534 and 4200000000 - 4294967294.

Tags

Specified tags to help identify a AWS Direct Connect resource.

No tags associated with the resource

[Add tag](#)[Cancel](#)[Create Direct Connect gateway](#)

Next we'll associate the **Transit gateway (TGW)** with the **Direct Connect gateway** to establish connectivity between the AWS environment and the on-premises environment.

4. Click **Direct Connect gateways** from the side menu. Click on the ID of previously created Direct Connect gateway.
5. Once on the Direct Connect gateway detail page, click on **Gateway Association** tab and further click on **Associate gateway**.
6. Choose the TGW (**Workshop-TGW**) already created as part of the workshop
7. In the **Allowed Prefixes** textbox, we'll include the CIDR of our VPC that we want to advertise back to on-premises environment. Put **10.1.0.0/24** into this field.
8. Click on **Associate gateway** to complete the step

Associate gateway

Association settings

Gateway

The gateway you want to associate with the new Direct Connect gateway.

Workshop-TGW
tgw-Odaa30cea3f5a6163 us-east-1

You can only associate gateways of the same type to a Direct Connect gateway. Transit gateways and virtual private gateways attached to a VPC are available for selection.

Allowed prefixes

List of prefixes you want to originate on the Direct Connect gateway and be advertised to the on-premises network.

10.1.0.0/24

Specify up to 200 prefixes, each prefix separated by a comma. Or, put each prefix on separate lines.

[Cancel](#)

[Associate gateway](#)

9. You will notice that the association is now listed as **associating**. It does take several minutes to associate the TGW to the Direct Connect Gateway. You can proceed without waiting for the association to complete.

Order Direct Connect Hosted Connection

In this part, i am going to set up a AWS Direct Connect Hosted Connection.

To order hosted connections, you place an order directly with the partner rather than via AWS. The ordering process is specific to each partner. In this case, i will order via a web portal.

1. Open the **PortalUrl** link from the Workshop Studio [dashboard](#) event outputs in a new browser window.
2. Login using the **PortalUsername** and **PortalPassword** outputs from the event dashboard.

dxgl.aws.com/login

Login

Username

QfhUB2m6Mlwa

Password

.....

[Sign in](#)

3. Click on **Order Direct Connect** from the left side menu.

Enter the connection details in the order form, then click **Submit Order**.

- **Connection Name:** DX
- **AWS Account Id:** Use the AwsAccountId value from the Workshop Studio [dashboard](#) event outputs
- **Router Interface (A-End):** router (eth1)
- **DX Location (B-End):** Choose any available location

dxgl.aws.com/order

Connection Name
Give your connection a name to help identify it in the AWS Console.
DX

AWS Account ID
Available from the account menu in the top right corner of the AWS Console.
727915193401

Speed
What bandwidth do you need for this connection?
50Mbps

Router Interface (A-End)
Which on-premises device and interface should this connection be delivered to?
router (eth1)

DX Location (B-End)
Which Direct Connect location should this connection be ordered at?
NEXTDC M1 port 1, Melbourne, Australia

VLAN
Which VLAN ID would you like to use for this connection?
☒ Auto Select VLAN

Cancel Submit Order

Connections

[Order New Connection](#)

Connection Name	Connection ID	Bandwidth	VLAN	AWS Account ID	Interface	Actions
DX	dxcon-fhbjde00	50Mbps	2411	727915193401	router	Delete

Accept the Hosted Connection

6. Open the **AWS Direct Connect Console**. Click **Connections** from the navigation menu. You will see that there is a **Connection** with the status **Ordering**. This means that the interface is waiting to be accepted for use by us. Click on the **Connection ID**.

Connections (1 of 1) Last updated March 12, 2026, 11:51 (UTC-05:00) [View details](#) [Edit](#) [Delete](#) [Create connection](#)

Q Search connections

☒	ID	Name	Region	Location	Bandwidth	State
☒	dxcon-fhbjde00	DX	ap-southeast-2	NEXTDC M1, M...	50Mbps	ordering

7. Click on the **Accept** button.

DXCON-FHBJDE00

AcceptDelete

General configuration

Connection ID dxcon-fhbjde00	State ordering	Location NEXTDC M1, Melbourne, AUS	Jumbo frame capable true
Connection name DX	Port speed 50Mbps	AWS logical device NDCM1-3rw6q0d7h6lbr	VLAN 2411
AWS account	Region	LOA issued at	Partner name

Create Transit Virtual Interface

Next we'll create a **Transit virtual interface (Transit VIF)**. After accepting the Direct Connect Hosted Connection, the screen will refresh and the details pane will show some additional tabs.

1. In the **Virtual Interfaces** detail pane, click **Create virtual interface**.

Virtual interfacesMonitoringTags

Virtual interfaces

Last updated
March 12, 2026, 11:56 (UTC-05:00)

View detailsEditDeleteActions

Create virtual interface

Search virtual interfaces

< 1 >

IDNameStateRegionConnecti...VLANType

No virtual interfaces
No virtual interfaces to display.
Create virtual interface

2. Specify the following option settings for the new Interface:
 - **Virtual interface type:** Transit
 - **Virtual interface name:** Transit VIF
 - **Connection:** leave as default
 - **Virtual interface owner:** leave as default
 - **Direct Connect gateway:** choose the Direct Connect Gateway from the drop down
 - **VLAN:** leave as default - if this isn't auto-populated, please use the same VLAN id as in the ordering portal
 - **BGP ASN:** 64513 (or choose your own, however this must be different to the ASN specified on the DX Gateway)

Transit virtual interface settings

Virtual interface name

A name to help you identify the new virtual interface.

My Transit VIF

Name must contain no more than 100 characters. Valid characters are a-z, 0-9, and hyphens (-).

Connection

The physical connection on which the new virtual interface will be provisioned.

DX

Virtual interface owner

The account that will own the virtual interface.

☒ My AWS account

☐ Another AWS account

Direct Connect gateway

The Direct Connect gateway to which the new virtual interface will be attached.

My Direct Connect Gateway

Virtual Local Area Network (VLAN)

The Virtual Local Area Network number for the new virtual interface.

2411

Valid ranges are 1 - 4094

BGP ASN

The Border Gateway Protocol (BGP) Autonomous System Number (ASN) of your on-premises router for the new virtual interface.

64513

Valid ranges are 1 - 4294967294.

3. Click **Additional Settings**

There are often times where you'd like to specify which BGP peer IP addresses to use. If you do not fill in these details, they will be automatically generated for you. If you'd like to follow along in this tutorial, use the following information:

- **Your router peer ip:** 169.254.100.1/30
- **Amazon router peer ip:** 169.254.100.2/30
- **BGP authentication key:** NTA308Password (or choose your own)
- **Leave all other options as default**
- **Click Create virtual interface**

▼ **Additional settings**

Address family - optional

Determines whether the virtual interface is created with an IPv4 or IPv6 peering.

☒ IPv4

☐ IPv6

Your router peer ip - optional

The BGP peer IP configured on your endpoint.

169.254.100.1/30

Amazon router peer IP - optional

The BGP peer IP configured on the AWS endpoint.

169.254.100.2/30

BGP authentication key - optional

The password that will be used to authenticate the BGP session.

NTA308Password

Jumbo MTU (MTU size 8500) - optional

Allow MTU size of 8500 on virtual interface.

☐ Enabled

Allow MTU size of 8500 on virtual interface.

Enable PIM - optional

Click the ID of the Virtual Interface. You will also see that the BGP status is **Down**. Next we'll configure the on-premises router and establish **BGP**.

Configure BGP Session

The Border Gateway Protocol, commonly known as BGP, is a method of exchanging routes between networks. BGP is a requirement to use Direct Connect. We will configure the on-premises router with the AWS Direct Connect Gateway.

1. Open the Consoles link from the Workshop Studio [dashboard](#) event outputs in a new browser window and choose the **router** device by pressing 2 on your Keyboard

The router we have for this workshop has 3 network interfaces.

- **eth0**: Internet
 - **eth1**: Direct Connect
 - **eth2**: LAN
2. Type **show interfaces** and press [Enter]. You'll see the following list of interfaces. eth1 is the interface we will be configuring for Direct Connect.

```
Connecting to router...
admin@router:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           100.127.255.40/24  u/u  Internet via ISP
eth1           -                u/u  Direct Connect via partner
eth2           10.0.0.1/24      u/u  On-premises LAN
lo             127.0.0.1/8      u/u
:::1/128
admin@router:~$
```

3. At the router terminal type **configure** and press **[Enter]** to enter configuration mode. You'll now see a prompt with a #

4. Next, we'll set the IP address for the sub-interface on eth1. Replace the **VLANID** with the same VLAN ID that was used when ordering the Direct Connect. Run the following commands to set the sub-interface IP address and commit the configuration

5. Type **run show interfaces** and press **[Enter]**. You can now see that we have a sub-interface on our eth1 interface with our VLAN and IP Address.

```

admin@router# run show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           100.127.255.40/24  u/u  Internet via ISP
eth1           -                u/u  Direct Connect via partner
eth1.2411      169.254.10.1/30   u/u
               169.254.100.1/30
eth2           10.0.0.1/24      u/u  On-premises LAN
lo             127.0.0.1/8      u/u
               ::1/128
[edit]
admin@router#

```

6.

Type the following commands. These commands will set up BGP, including specifying our local AS Number, advertising our local network **10.0.0.0/24**, and defining the Direct Connect Gateway as the BGP neighbor. You can copy and paste for convenience.

```

[edit]
admin@router# set protocols bgp 64513 parameters router-id 169.254.100.1
[edit]
admin@router# set protocols bgp 64513 address-family ipv4-unicast network 10.0.0.0/24
[edit]
admin@router# set protocols bgp 64513 neighbor 169.254.100.2 remote-as 64512
[edit]
admin@router# set protocols bgp 64513 neighbor 169.254.100.2 password NTA308Password
[edit]
admin@router# set protocols bgp 64513 neighbor 169.254.100.2 address-family ipv4-unicast
[edit]
admin@router# set protocols bgp 64513 neighbor 169.254.100.2 solo
[edit]
admin@router# set protocols bgp 64513 neighbor 169.254.100.2 bfd
[edit]
admin@router# set protocols bfd peer 169.254.100.2
[edit]
admin@router# commit
[edit]
admin@router#

```

7. The **commit** command will activate the changes

8. Type **exit** and press **[Enter]** to exit out of configuration mode.

9. Type **show ip bgp summary** to verify that the connection and the current status. You should be able to see a neighbor listed, with Up/Down displaying a time that the connection has been established.

```

admin@router:~$ show ip bgp summary

IPv4 Unicast Summary:
BGP router identifier 169.254.100.1, local AS number 64513 vrf-id 0
BGP table version 2
RIB entries 3, using 576 bytes of memory
Peers 1, using 21 KiB of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  PfxSnt
169.254.100.2  4      64513    9         8        0    0    0 00:02:27      1         1

Total number of neighbors 1
admin@router:~$

```

10. Type **show ip bgp ipv4 unicast** detail and press **[Enter]** if you would like to see the routes that are being received.

Test Direct Connect Connectivity

To verify the connectivity between AWS and Simulated On-premises via Direct Connect (DX), we are going to use Bastion-Host EC2 instance to access On-premise web server behind the On-premises router we configured using BGP in previous step. The Web server IP is **10.0.0.2**.

Before we test the connectivity, let's make the final step of adding the route for On-premises LAN **10.0.0.0/24** in Central VPC route table pointing to Transit Gateway.

Add LAN router in Central VPC route table

1. Open the **VPC Console**, and choose **Route Tables** from the side menu.
2. Select the check box for **Central-Private-RT** route table, scroll down select the **Routes** tab and click on **Edit routes**.

The screenshot shows the AWS VPC Console interface. On the left, the 'Virtual private cloud' sidebar is visible, with 'Route tables' selected. The main panel displays a list of route tables. The 'Central-Private-RT' route table is selected, and the 'Routes' tab is active. Below the route table list, the 'Routes (3)' section is shown, with a table of routes. The route for '2600:1f18:4e78:b900::/56' is highlighted, showing a status of 'Active' and 'Propagated' set to 'No'.

Destination	Target	Status	Propagated	Route Origin
2600:1f18:4e78:b900::/56	local	Active	No	Create Route Table

3. Add a route to On-premises LAN **10.0.0.0/24** in the Central VPC Private Route Table using a **Destination of Transit Gateway (TGW Central VPC Attachment)**.
4. Click on **Save routes**.

Edit routes

Destination	Target	Status	Propagated	Route Origin	
2600:1f18:4e78:b900::/56	local	Active	No	CreateRouteTable	
10.1.0.0/24	local	Active	No	CreateRouteTable	
10.2.0.0/24	Transit Gateway	Active	No	CreateRoute	Remove
10.0.0.0/24	Transit Gateway	-	No	CreateRoute	Remove

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

Test Connectivity

1. Open the **EC2 Console**, and choose **Instances** from the side menu.
2. Click on the **Bastion-Host** instance and click the **Connect** button.

Instances (1/5) [Info](#) [Refresh](#) [Connect](#)

Find Instance by attribute or tag (case-sensitive) [All states](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm
☐	OnPrem-VPN-...	i-0f9ee0fd3095f3156	Running	t2.micro	2/2 checks passed	
☐	OpenSWAN-C...	i-09162eb40a5dbfc4e	Running	t3.micro	3/3 checks passed	
☐	VPC-A-Instance	i-0f7aba63c301140a3	Running	t2.micro	2/2 checks passed	
☒	Bastion-Host	i-0e8d5a9d721587b5c	Running	t3.micro	3/3 checks passed	
☐	OnPrem-DX-R...	i-01e5a3b2f23dc62b8	Running	m5.large	3/3 checks passed	

3. In the Connect to instance page, select the **Session Manager** tab and click on **Connect** button.

EC2 Instance Connect **SSM Session Manager** SSH client EC2 serial console

SSM agent status

Ping status: **Online** Last ping time: Thu Mar 12 2026 15:11:06 GMT-0500 (Central Daylight Time) Session Manager connection status: **Connected** Agent version: 3.3.3598.0

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

[Cancel](#) [Connect](#)

4. Once the session open, do `ping 10.0.0.2` and `curl http://10.0.0.2` to confirm connectivity via Direct Connect works.

```

sh-4.2$ ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=61 time=422 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=61 time=421 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=61 time=421 ms
^C
--- 10.0.0.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 421.334/421.626/422.029/0.606 ms
sh-4.2$ curl http://10.0.0.2
<html>
  <head>
    <title>Home Page - re:invent 2025</title>
  </head>
  <body>
    Welcome to re:invent 2025
  </body>
</html>
sh-4.2$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 10.1.0.93 netmask 255.255.255.128 broadcast 10.1.0.127
    inet6 fe80::cd3:76ff:fe27:9eb3 prefixlen 64 scopeid 0x20<link>
    inet6 2600:1f18:4e78:b900:1acc:d0c8:6b1f:6260 prefixlen 128 scopeid 0x0<global>
    ether 0e:d3:76:27:9e:b3 txqueuelen 1000 (Ethernet)
    RX packets 79125 bytes 11037834 (10.5 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 67228 bytes 8766001 (8.3 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 64 bytes 5232 (5.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 64 bytes 5232 (5.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```

In this lab, I learned how to set up Direct Connections using public and private VIFs. I also learned what VIFs are and how they work to direct traffic flow. I concluded by testing my work.